

This is due by August 8.

Problem 1. Let k be a field and $f(x) \in k[x]$ be a polynomial.

1. Suppose $\deg f = 2$ or 3 . Show f is irreducible iff f has no roots in k .
2. If k is not a field but a domain, does the previous statement hold?
3. Give an example of a field k and a degree 4 polynomial $f(x)$ that has no roots in k such that $f(x)$ is not irreducible.

proof of part 2. (Fred.) Consider $(2x - 1)^2$ in $\mathbb{Z}[x]$.

(Tianshuo.) Consider $(2x + 1)(2x + 3)$ in $\mathbb{Z}[x]$.

(Nikoli.) Consider $2(x^2 + 1)$ in $\mathbb{Z}[x]$

□

proof of part 3. (Jenny.) Consider $x^4 + 1 = (x^2 - \sqrt{2}x + 1)(x^2 + \sqrt{2}x + 1)$ in $\mathbb{R}[x]$.

(Delano.) Consider $x^4 + x^2 + 1 = (x^2 + x + 1)^2$ in $\mathbb{Z}_2[x]$.

(Albert, Marcial.) Consider $(x^2 + 1)^2$ in $\mathbb{R}[x]$.

□

Problem 2. Let R be a domain.

1. Show an element $0 \neq r \in R$ is irreducible iff (r) is maximal among the principal ideals (i.e, if $(r) \subseteq (a)$ for some $a \in R$, then $(a) = (r)$ or $(a) = R$.)
2. Recall $\mathbb{Z}[\sqrt{-2}]$ is a ED. Give an example of a maximal ideal.

proof of part 2. Note $1 + \sqrt{2}$ has norm 3 which is a prime in $\mathbb{Z}$. If $1 + \sqrt{2} = ab$ in $\mathbb{Z}[\sqrt{2}]$, then $N(a)N(b) = 3$ which implies one of them must be of ± 1 . But any norm ± 1 element will be a unit since $a^2 + 2b^2 = 1$ has only four solutions (and you may check they are all units). □

Problem 3. Let R be a commutative ring and consider $R[x]$.

1. If R is a field, then (x) is a maximal ideal. Is $(x + 1)$ a prime ideal?
2. If R is a domain, then (x) is a prime ideal. Is (x) a maximal ideal?
3. Is $(x + 1)$ a prime ideal in $\mathbb{Z}[x]$?

- Proof.* 1. Note x is irreducible so (x) is maximal among the principle ideals. But $R[x]$ is PID when R is a field. Also note $x + 1$ is irreducible so it is prime and $(x + 1)$ is prime because $R[x]$ is UFD when R is a field.
2. Let $fg \in (x)$. Then the constant term of f and g must be zero because any element in (x) has zero constant term. But R is a domain. So either f or g has zero constant term and hence lies in (x) . No (x) is not a maximal ideal because $(x) \subseteq (x, 2) \neq \mathbb{Z}[x]$.
3. Note $x + 1$ is irreducible in $\mathbb{Z}[x]$ and hence it is prime because $\mathbb{Z}[x]$ is UFD. Then $(x + 1)$ is prime ideal. $\square$

Problem 4. Let $f : R \rightarrow S$ be a ring homomorphism between commutative rings.

1. Let k be a field and $p \in k$. Show $ev_p : k[x] \rightarrow k$ has kernel $(x - p)$.
2. Consider ring homomorphism $\rho : \mathbb{R}[x] \rightarrow \mathbb{C}$ determined by $x \mapsto i$. Show the kernel is $(x^2 + 1)$.

- Proof.* 1. Clearly $(x - p) \subseteq \ker ev_p$. But $(x - p)$ is maximal. So $\ker ev_p = (x - p)$ or $k[x]$. But $\ker ev_p \neq k[x]$ for 1 is not in $\ker ev_p$.
2. Similar argument for $(x^2 + 1)$ is maximal as well. $\square$

Problem 5. Let k be a field. Consider the polynomial ring over the polynomial ring over a field

$$k[x][y] = \left\{ \sum_{i=0}^n f_i(x)y^i : f_i(x) \in k[x], n \in \mathbb{N} \right\}.$$

This is really just $k[x, y] = \{ \sum_{i=0}^n f_i(x)y^i : f_i(x) \in k[x] \}$. Explain why $\mathbb{Z}[x]$ is not an ED by showing (x, y) is not a principle ideal.